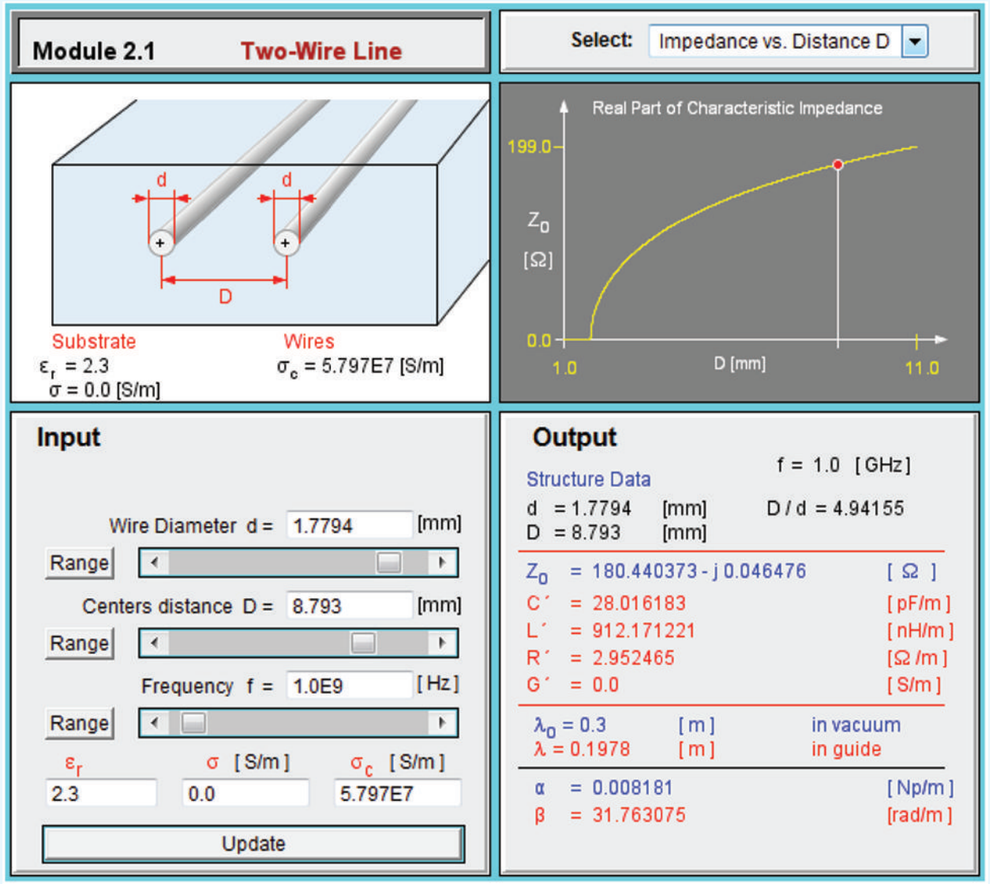
Module 2.1 Two-Wire Line The input data specifies the geometric and electric parameters of a two-wire transmission line. The output includes the calculated values for the line parameters, characteristic impedance Z_0, and attenuation and phase constants, as well as plots of Z_0 as a function of d and D.



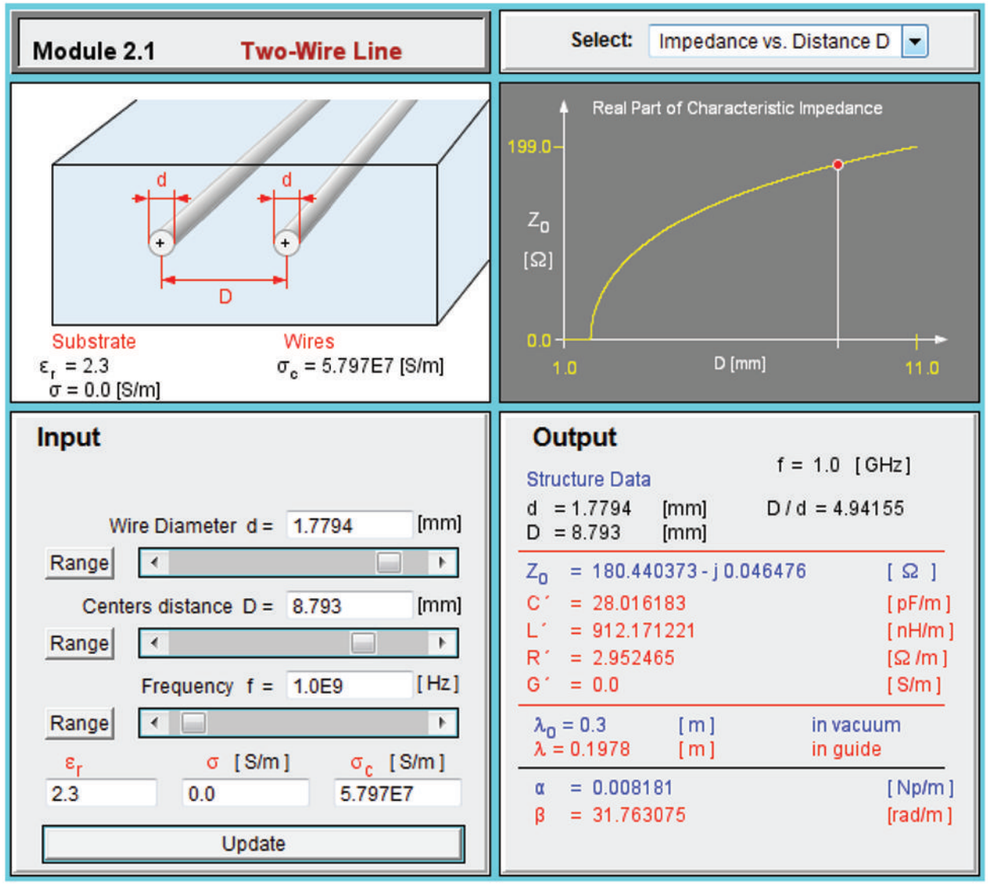
to why one of the ratios is the negative of the other. The explanation, which is available in more detail in Chapter 7, is based on a directional rule that specifies the relationships between the directions of the electric and magnetic fields of a TEM wave and its direction of propagation. On a transmission line, the voltage is related to the electric field **E** and the current is related to the magnetic field **H**. To satisfy the directional rule, reversing the direction of propagation requires reversal of the direction (or polarity) of I relative to V. Hence, $V_0^-/I_0^- = -V_0^+/I_0^+$.

In terms of Z_0, Eq. (2.27) can be cast in the form

$$\tilde{I}(z) = \frac{V_0^+}{Z_0} e^{-\gamma z} - \frac{V_0^-}{Z_0} e^{\gamma z}. \tag{2.30}$$

56 CHAPTER 2 TRANSMISSION LINES

模块 2.1 两线传输线 输入数据指定两线传输线的几何和电气参数。输出包括线路参数、特性阻抗 Z_0、衰减和相位常数的计算值，以及作为 d 和 D 函数的 Z_0 绘图。

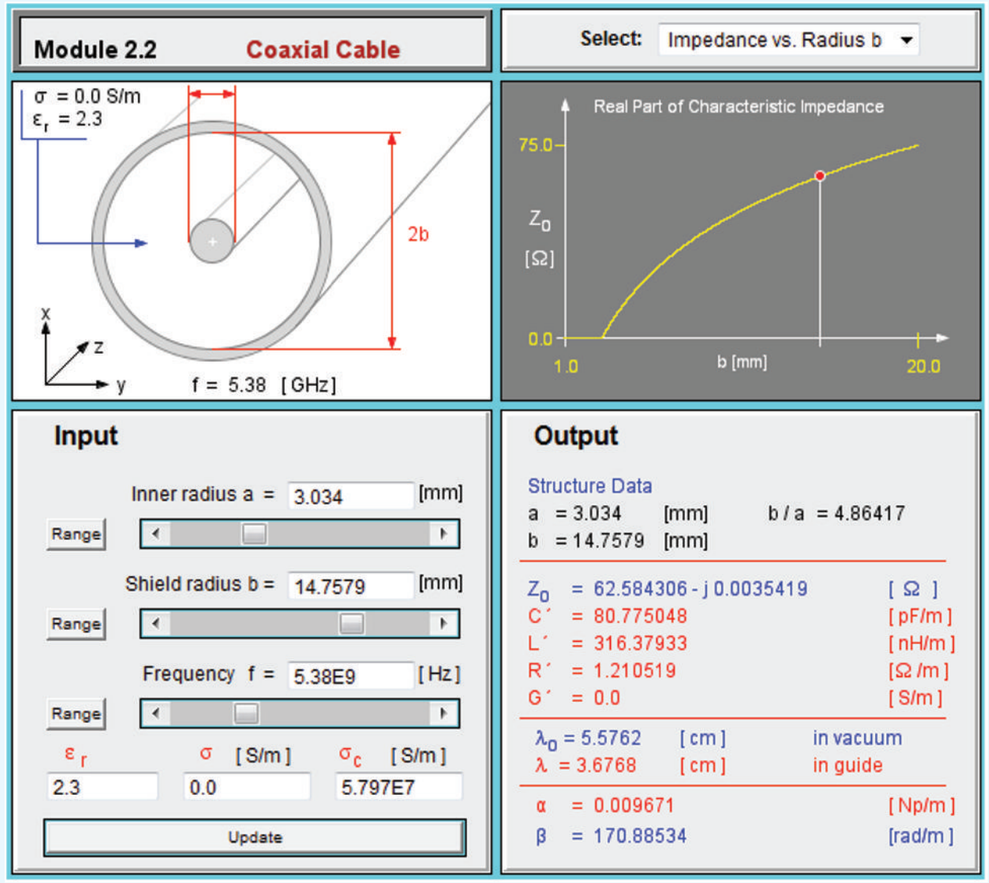


为什么其中一个比率是另一个比率的负数。第 7 章中有更详细的解释，它基于方向规则，该规则指定了 TEM 波的电场和磁场方向及其传播方向之间的关系。在传输线上，电压与电场 E 相关，电流与磁场 H 相关。满足根据方向规则，反转传播方向需要反转 I 相对于 V 的方向（或极性）。因此，$V_0^-/我^- = -V_0^+/我^+$。

就 Z_0 而言，方程：(2.27) 可以转换为以下形式

$$\tilde{I}(z) = \frac{V_0^+}{Z_0} e^{-\gamma z} - \frac{V_0^-}{Z_0} e^{\gamma z}. \tag{2.30}$$

Module 2.2 Coaxial Line Except for changing the geometric parameters to those of a coaxial transmission line, this module offers the same output information as Module 2.1.



According to Eq. (2.29), the characteristic impedance Z_0 is determined by the angular frequency ω of the wave traveling along the line and the four line parameters (R', L', G', and C'). These, in turn, are determined by the line geometry and its constitutive parameters. Consequently, the combination of Eqs. (2.26a) and (2.30) now contains only two unknowns, namely V_0^+ and V_0^-, as opposed to four.

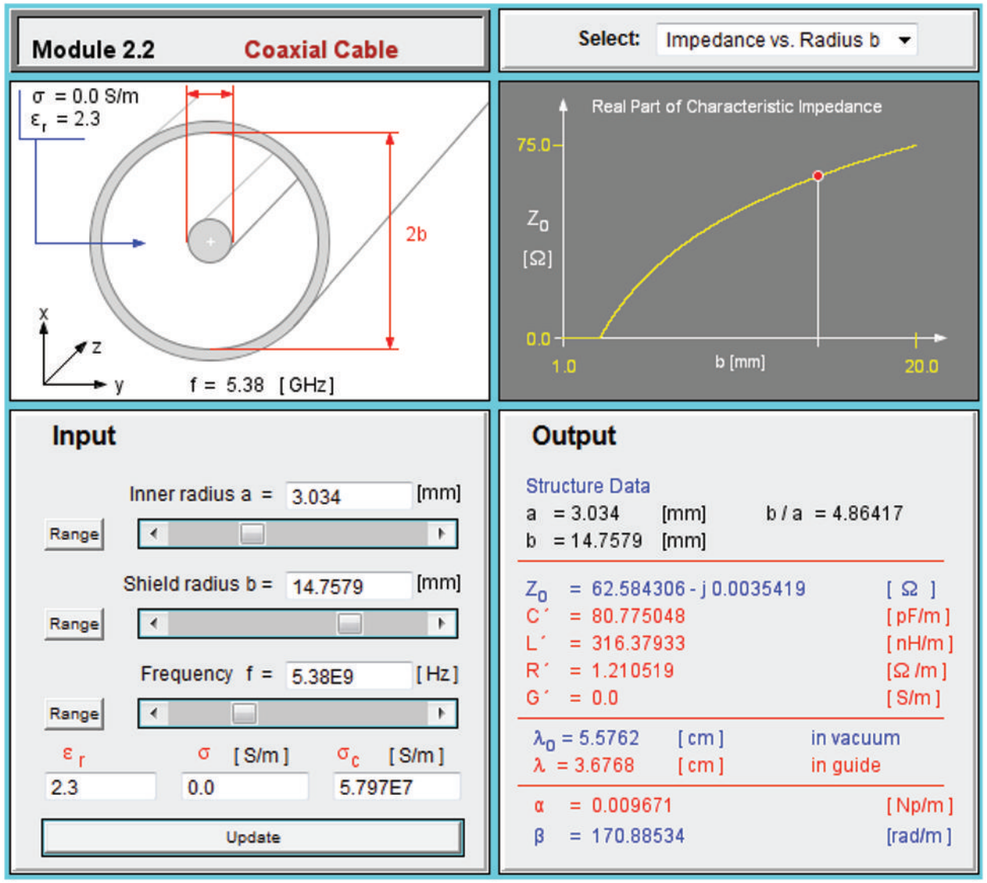
2-4.2 Time-Domain Solution for $v(z,t)$

In later sections, we apply boundary conditions at the source and load ends of the transmission line to obtain expressions for the remaining wave amplitudes V_0^+ and V_0^-. In general, each is a complex quantity characterized by a magnitude and a phase angle:

$$V_0^+ = |V_0^+|e^{j\phi^+}, \tag{2.31a}$$

2-4 WAVE PROPAGATION ON A TRANSMISSION LINE 57

模块2.2 同轴线 除了将几何参数更改为同轴传输线的几何参数外，该模块提供与模块2.1 相同的输出信息。



根据方程。根据方程（2.29），特性阻抗 Z_0 由沿线传播的波的角频率 ω 和四个线参数（R′、L′、G′ 和 C′）确定。这些又由线的几何形状及其本构参数决定。因此，等式的组合。 (2.26a) 和 (2.30) 现在仅包含两个未知数，即 V_0^+ 和 V_0^-，而不是四个。

2-4.2 υ(z, t) 的时域解

在后面的部分中，我们在传输线的源端和负载端应用边界条件，以获得剩余波幅 V_0^+ 和 V_0^- 的表达式。一般来说，每个都是一个由幅度和相位角表征的复量：

$$V_0^+ = |V_0^+|e^{j\phi^+}, \tag{2.31a}$$

$$V_0^- = |V_0^-|e^{j\phi^-}. \tag{2.31b}$$

After substituting these definitions in Eq. (2.26a) and using Eq. (2.24) to decompose γ into its real and imaginary parts, we can convert back to the time domain to obtain an expression for $v(z,t)$, the instantaneous voltage on the line:

$$\begin{aligned} v(z,t) &= \Re\mathrm{e}(\widetilde{V}(z)e^{j\omega t}) \\ &= \Re\mathrm{e}\left[\left(V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}\right) e^{j\omega t}\right] \\ &= \Re\mathrm{e}[|V_0^+|e^{j\phi^+} e^{j\omega t} e^{-(\alpha+j\beta)z} + |V_0^-|e^{j\phi^-} e^{j\omega t} e^{(\alpha+j\beta)z}]. \end{aligned}$$

The final expression for $v(z,t)$ is then given by

$$\begin{aligned} v(z,t) = {} & |V_0^+|e^{-\alpha z}\cos(\omega t - \beta z + \phi^+) \\ & + |V_0^-|e^{\alpha z}\cos(\omega t + \beta z + \phi^-). \end{aligned} \tag{2.32}$$

From our review of waves in Section 1-4, we recognize the first term in Eq. (2.32) as a wave traveling in the $+z$ direction (the coefficients of t and z have opposite signs) and the second term as a wave traveling in the $-z$ direction (the coefficients of t and z are both positive). Both waves propagate with a phase velocity u_p given by Eq. (1.30):

$$u_p = f\lambda = \frac{\omega}{\beta}. \tag{2.33}$$

Because the wave is *guided* by the transmission line, λ often is called the ***guide wavelength***. The factor $e^{-\alpha z}$ accounts for the attenuation of the $+z$ propagating wave, and the factor $e^{\alpha z}$ accounts for the attenuation of the $-z$ propagating wave.

▶ The presence of two waves on the line propagating in opposite directions produces a ***standing wave***. ◀

To gain a physical understanding of what that means, we later provide a thorough treatment of standing waves in Section 2-6.2.

2-4.3 Time-Domain Solution for $i(z,t)$

In the phasor domain, the expression for $\widetilde{I}(z)$ given by Eq. (2.30) is similar to that for $\widetilde{V}(z)$, except that the current amplitudes are divided by the characteristic impedance of the transmission line, Z_0, and the current wave traveling in the $-z$ direction has a minus sign (second term in Eq. (2.30)). Since Z_0 is, in general, a complex quantity (see Eq. (2.29)), we need to express it as

$$Z_0 = |Z_0|e^{j\phi_z}. \tag{2.34}$$

To obtain an expression for $i(z,t)$, we insert the expressions given by Eqs. (2.31) and (2.34) into Eq. (2.30), multiply both terms by $e^{j\omega t}$, and then take the real part:

$$\begin{aligned} i(z,t) = \Re\mathrm{e}(\widetilde{I}(z)\, e^{j\omega t}) = \Re\mathrm{e}\Bigg[& \frac{|V_0^+|}{|Z_0|} e^{j\phi^+} e^{-j\phi_z} e^{j\omega t} e^{-(\alpha+j\beta z)} \\ & - \frac{|V_0^-|}{|Z_0|} e^{j\phi^-} e^{-j\phi_z} e^{j\omega t} e^{(\alpha+j\beta z)}\Bigg], \end{aligned}$$

which yields

$$\begin{aligned} i(z,t) = {} & \frac{|V_0^+|}{|Z_0|} e^{-\alpha z}\cos(\omega t - \beta z + \phi^+ - \phi_z) \\ & - \frac{|V_0^-|}{|Z_0|} e^{\alpha z}\cos(\omega t + \beta z + \phi^- - \phi_z). \end{aligned} \tag{2.35}$$

Comparison of the expression for $i(z,t)$ with that given by Eq. (2.32) for $v(z,t)$ reveals that:

(a) The amplitudes of the current waves are reduced by $|Z_0|$ when compared with the amplitudes of the voltage waves.

(b) The amplitude of the second term in Eq. (2.35)—corresponding to the current wave traveling in the negative z direction—has a minus sign.

(c) The reference phase angle in the expression for $i(z,t)$ has an extra component, namely $-\phi_z$, when compared with the reference phase angle of $v(z,t)$.

Example 2-1: Air Line

An ***air line*** is a transmission line in which air separates the two conductors, which renders $G' = 0$ because $\sigma = 0$. In addition, assume that the conductors are made of a material with high conductivity so that $R' \approx 0$. For an air line with a characteristic impedance of 50 Ω and a phase constant of 20 rad/m at 700 MHz, find the line inductance L' and the line capacitance C'.

Solution: The following quantities are given:

$$\begin{aligned} Z_0 &= 50\ \Omega, \\ \beta &= 20\ \text{rad/m}, \\ f &= 700\ \text{MHz} = 7\times 10^8\ \text{Hz}. \end{aligned}$$

With $R' = G' = 0$, Eqs. (2.25b) and (2.29) reduce to

$$\beta = \Im\mathrm{m}\left[\sqrt{(j\omega L')(j\omega C')}\right] = \Im\mathrm{m}\left(j\omega\sqrt{L'C'}\right) = \omega\sqrt{L'C'}\,,$$

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$$V_0^- = |V_0^-|e^{j\phi^-}. \tag{2.31b}$$

将这些定义代入等式后： （2.26a）并使用方程。(2.24) 将 γ 分解为实部和虚部，我们可以转换回时域，得到 υ(z ,t) 的表达式，即线路上的瞬时电压：

$$\begin{aligned} v(z,t) &= \Re\mathrm{e}(\widetilde{V}(z)e^{j\omega t}) \\ &= \Re\mathrm{e}\left[\left(V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}\right) e^{j\omega t}\right] \\ &= \Re\mathrm{e}[|V_0^+|e^{j\phi^+} e^{j\omega t} e^{-(\alpha+j\beta)z} + |V_0^-|e^{j\phi^-} e^{j\omega t} e^{(\alpha+j\beta)z}]. \end{aligned}$$

υ(z,t) 的最终表达式由下式给出

$$\begin{aligned} v(z,t) = {} & |V_0^+|e^{-\alpha z}\cos(\omega t - \beta z + \phi^+) \\ & + |V_0^-|e^{\alpha z}\cos(\omega t + \beta z + \phi^-). \end{aligned} \tag{2.32}$$

从我们对第 1-4 节中的波的回顾中，我们认识到方程中的第一项。 (2.32) 为沿 +z 方向传播的波（t 和 z 的系数具有相反的符号），第二项为沿 −z 方向传播的波（t 和 z 的系数均为正）。两个波都以等式给出的相速度 u_p 传播。 （1.30）：

$$u_p = f\lambda = \frac{\omega}{\beta}. \tag{2.33}$$

由于波是由传输线引导的，因此λ通常称为波导波长。因子 $e^{-\alpha z}$ 说明了 +z 传播波的衰减，因子 $e^{\alpha z}$ 说明了 −z 传播波的衰减。

▶ 线上存在沿相反方向传播的两个波会产生驻波。 ◀

为了从物理上理解这意味着什么，我们稍后在 2-6.2 节中对驻波进行彻底的处理。

2-4.3 i(z,t)的时域解

在相量域中，~I(z) 的表达式由式（1）给出。方程（2.30 ）与~V（z）类似，不同之处在于电流幅值除以传输线的特性阻抗Z_0，并且沿−z方向传播的电流波具有负号（方程（2.30）中的第二项）。由于 Z_0 通常是一个复数（参见方程（2.29）），我们需要将其表示为

$$Z_0 = |Z_0|e^{j\phi_z}. \tag{2.34}$$

为了获得 i(z,t) 的表达式，我们插入方程 1 给出的表达式 。将（2.31）和（2.34）代入方程。(2.30)，将两项乘以 $e^{j\omega t}$，然后取实部：

$$\begin{aligned} i(z,t) = \Re\mathrm{e}(\widetilde{I}(z)\, e^{j\omega t}) = \Re\mathrm{e}\Bigg[& \frac{|V_0^+|}{|Z_0|} e^{j\phi^+} e^{-j\phi_z} e^{j\omega t} e^{-(\alpha+j\beta z)} \\ & - \frac{|V_0^-|}{|Z_0|} e^{j\phi^-} e^{-j\phi_z} e^{j\omega t} e^{(\alpha+j\beta z)}\Bigg], \end{aligned}$$

这产生

$$\begin{aligned} i(z,t) = {} & \frac{|V_0^+|}{|Z_0|} e^{-\alpha z}\cos(\omega t - \beta z + \phi^+ - \phi_z) \\ & - \frac{|V_0^-|}{|Z_0|} e^{\alpha z}\cos(\omega t + \beta z + \phi^- - \phi_z). \end{aligned} \tag{2.35}$$

i(z,t) 的表达式与方程给出的表达式的比较。 (2.32) 对于 υ(z,t) 表明：

(a) 与电压波的振幅相比，电流波的振幅减少了$|Z_0|$。 (b) 等式中第二项的幅度。 (2.35)——对应于在负 z 方向上传播的电流波——有一个负号。 (c) 与υ(z,t)的参考相位角相比，i(z,t)表达式中的参考相位角多了一个分量，即 $-\phi_z$。

示例 2-1：航空线路

空气线路是一种传输线路，其中空气将两个导体分开，这使得 $G' = 0$ 因为 $\sigma = 0$。此外，假设导体由高电导率材料制成，因此 $R' \approx 0$。对于特性阻抗为 50 Ω 且在 700 MHz 时相位常数为 20 rad/m 的空气线路，求出线路电感 L′ 和线路电感电容C ′。

解：给出下列数量：

$$\begin{aligned} Z_0 &= 50\ \Omega, \\ \beta &= 20\ \text{rad/m}, \\ f &= 700\ \text{MHz} = 7\times 10^8\ \text{Hz}. \end{aligned}$$

当 $R' = G' = 0$ 时，方程(2.25b) 和 (2.29) 减少为

$$\beta = \Im\mathrm{m}\left[\sqrt{(j\omega L')(j\omega C')}\right] = \Im\mathrm{m}\left(j\omega\sqrt{L'C'}\right) = \omega\sqrt{L'C'}\,,$$

$$Z_0 = \sqrt{\frac{j\omega L'}{j\omega C'}} = \sqrt{\frac{L'}{C'}}.$$

The ratio of β to Z_0 is $\beta/Z_0 = \omega C'$, or

$$C' = \frac{\beta}{\omega Z_0} = \frac{20}{2\pi \times 7 \times 10^8 \times 50} = 9.09 \times 10^{-11} \text{ (F/m)}$$
$$= 90.9 \text{ (pF/m)}.$$

From $Z_0 = \sqrt{L'/C'}$, it follows that

$$L' = Z_0^2 C' = (50)^2 \times 90.9 \times 10^{-12} = 2.27 \times 10^{-7} \text{ (H/m)}$$
$$= 227 \text{ (nH/m)}.$$

Exercise 2-3: Verify that Eq. (2.26a) indeed provides a solution to the wave equation (2.21).

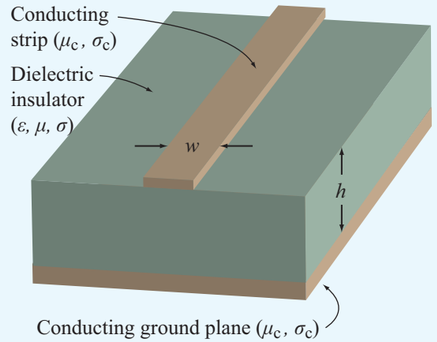
Answer: (See EM.)

Exercise 2-4: A two-wire air line has the following line parameters: $R' = 0.404$ (mΩ/m), $L' = 2.0$ (μH/m), $G' = 0$, and $C' = 5.56$ (pF/m). For operation at 5 kHz, determine (a) the attenuation constant α, (b) the phase constant β, (c) the phase velocity u_p, and (d) the characteristic impedance Z_0. (See EM.)

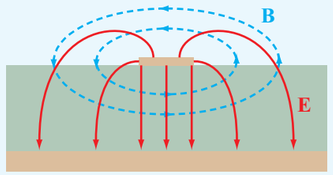
Answer: (a) $\alpha = 3.37 \times 10^{-7}$ (Np/m), (b) $\beta = 1.05 \times 10^{-4}$ (rad/m), (c) $u_p = 3.0 \times 10^8$ (m/s), (d) $Z_0 = (600 - j1.9)\ \Omega = 600\angle -0.18^\circ\ \Omega$.

2-5 The Lossless Microstrip Line

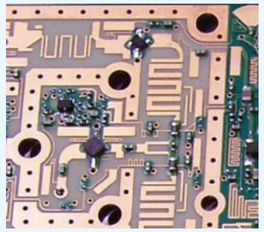
Because its geometry is well suited for fabrication on printed circuit boards, the microstrip line is the most common interconnect configuration used in RF and microwave circuits. It consists of a narrow, very thin strip of copper (or another good conductor) printed on a dielectric substrate overlaying a ground plane (**Fig. 2-10(a)**). The presence of charges of opposite polarity on its two conducting surfaces gives rise to electric field lines between them (**Fig. 2-10(b)**). Also, the flow of current through the conductors (when part of a closed circuit) generates magnetic field loops around them, as illustrated in **Fig. 2-10(b)** for the narrow strip. Even though the patterns of **E** and **B** are not everywhere perfectly orthogonal, they are approximately so in the region between the conductors, which is where the **E** and **B** fields are concentrated the most. Accordingly, the microstrip line is considered a ***quasi-TEM*** transmission line, which allows us to describe its voltages and currents in terms of the one-dimensional TEM model of Section 2-4, namely Eqs. (2.26) through (2.33).



(a) Longitudinal view



(b) Cross-sectional view with **E** and **B** field lines



(c) Microwave circuit

Figure 2-10 Microstrip line: (a) longitudinal view, (b) cross-sectional view, and (c) circuit example. (Courtesy of Prof. Gabriel Rebeiz, U. California at San Diego.)

The microstrip line has two geometric parameters, the ***width of the elevated strip***, w, and the ***thickness (height) of the dielectric layer***, h. We will ignore the thickness of the conducting strip because it has a negligible influence on the propagation properties of the microstrip line so long as the strip thickness is much smaller than the width w, which is almost

$$Z_0 = \sqrt{\frac{j\omega L'}{j\omega C'}} = \sqrt{\frac{L'}{C'}}.$$

β与Z_0的比率为$\beta/Z_0 = \omega C'$，或

$$C' = \frac{\beta}{\omega Z_0} = \frac{20}{2\pi \times 7 \times 10^8 \times 50} = 9.09 \times 10^{-11} \text{ (F/m)}$$
$$= 90.9 \text{ (pF/m)}.$$

从 $Z_0 = \sqrt{L'/C'}$ 可以得出

$$L' = Z_0^2 C' = (50)^2 \times 90.9 \times 10^{-12} = 2.27 \times 10^{-7} \text{ (H/m)}$$
$$= 227 \text{ (nH/m)}.$$

练习 2-3：验证等式。（2.26a）确实提供了波动方程（2.21）的解。

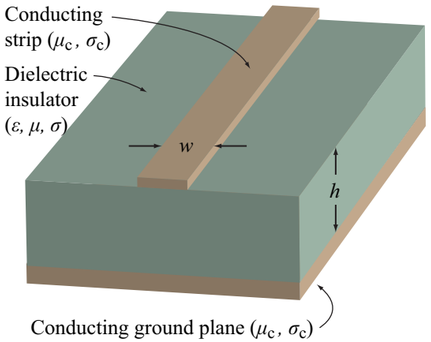
答案：（参见EM。）

练习 2-4：两线空气管路具有以下管路参数：$R' = 0.404$ (mΩ/m)、$L' = 2.0$ (μH/m)、$G' = 0$ 和 $C' = 5.56$ (pF/m)。对于工作频率为 5 kHz 的情况，确定 (a) 衰减常数 α、(b) 相位常数 β、(c) 相速度 u_p 和 (d) 特征阻抗 Z_0。（参见EM。）

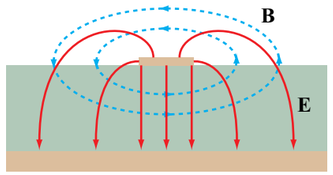
答案：(a) $\alpha = 3.37 \times 10^{-7}$ (Np/m)，(b) $\beta = 1.05 \times 10^{-4}$ (rad/m)，(c) $u_p = 3.0 \times 10^8$ (m/s)，(d) $Z_0 = (600 - j1.9)\ \Omega = 600 - 0.18^\circ\ \Omega$。

2-5 无损微带线

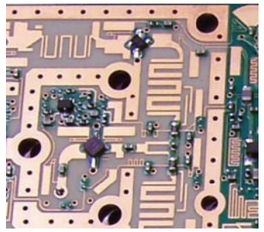
由于其几何形状非常适合在印刷电路板上制造，因此微带线是射频和微波电路中最常用的互连配置。它由一条窄且非常薄的铜条（或其他良导体）组成，印刷在覆盖接地层的电介质基板上（图 2-10(a)）。其两个导电表面上存在相反极性的电荷，从而在它们之间产生电场线（图2-10（b））。此外，流经导体（作为闭合电路的一部分时）的电流会在导体周围产生磁场环，如图 2-10(b) 中的窄带所示。尽管 E 和 B 的图案并非处处都完全正交，但它们在导体之间的区域大致如此，这是 E 和 B 场最集中的地方。因此，微带线被认为是准 TEM 传输线，这使我们能够根据第 2-4 节的一维 TEM 模型（即方程 1）来描述其电压和电流。（2.26）到（2.33）。



(a) Longitudinal view



(b) Cross-sectional view with **E** and **B** field lines



(c) Microwave circuit

图 2-10 微带线：(a) 纵向视图，(b) 横截面视图，(c) 电路示例。（由美国加州圣地亚哥的 Gabriel Rebeiz 教授提供。）

微带线有两个几何参数，即抬高带的宽度 w 和介电层的厚度（高度）h。我们将忽略导电带的厚度，因为只要导电带的厚度远小于宽度 w，它对微带线的传播特性的影响就可以忽略不计，宽度 w 几乎是

always the case in practice. Also, we assume the substrate material to be a perfect dielectric with $\sigma = 0$ and the metal strip and ground plane to be perfect conductors with $\sigma_c \approx \infty$. These two assumptions simplify the analysis considerably without incurring significant error. Finally, we set $\mu = \mu_0$, which is always true for the dielectric materials used in the fabrication of microstrip lines. These simplifications reduce the number of geometric and material parameters to three, namely w, h, and ε.

Electric field lines always start on the conductor carrying positive charges and end on the conductor carrying negative charges. For the coaxial, two-wire, and parallel-plate lines shown in the upper part of **Fig. 2-4**, the field lines are confined to the region between the conductors. A characteristic attribute of such transmission lines is that the phase velocity of a wave traveling along any one of them is given by

$$u_p = \frac{c}{\sqrt{\varepsilon_r}}, \tag{2.36a}$$

where c is the velocity of light in free space and ε_r is the relative permittivity of the dielectric medium between the conductors.

2-5.1 Effective Permittivity

In the microstrip line, even though most of the electric field lines connecting the strip to the ground plane do pass directly through the dielectric substrate, a few go through both the air region above the strip and the dielectric layer (**Fig. 2-10(b)**). This nonuniform mixture can be accounted for by defining an ***effective relative permittivity*** ε_{eff} such that the phase velocity is given by an expression that resembles Eq. (2.36a), namely

$$u_p = \frac{c}{\sqrt{\varepsilon_{eff}}}. \tag{2.36b}$$

Methods for calculating the propagation properties of the microstrip line are quite complicated and beyond the scope of this text. However, it is possible to use curve-fit approximations to rigorous solutions to arrive at the following set of expressions:*

$$\varepsilon_{eff} = \frac{\varepsilon_r + 1}{2} + \left(\frac{\varepsilon_r - 1}{2}\right)\left(1 + \frac{10}{s}\right)^{-xy}, \tag{2.37a}$$

where s is the ***width-to-thickness ratio***,

$$s = \frac{w}{h}, \tag{2.37b}$$

and x and y are intermediate variables given by

$$x = 0.56\left[\frac{\varepsilon_r - 0.9}{\varepsilon_r + 3}\right]^{0.05}, \tag{2.38a}$$

$$y = 1 + 0.02\ln\left(\frac{s^4 + 3.7\times10^{-4}s^2}{s^4 + 0.43}\right) + 0.05\ln(1 + 1.7\times10^{-4}s^3). \tag{2.38b}$$

2-5.2 Characteristic Impedance

The characteristic impedance of the microstrip line is given by

$$Z_0 = \frac{60}{\sqrt{\varepsilon_{eff}}}\ln\left\{\frac{6 + (2\pi - 6)e^{-t}}{s} + \sqrt{1 + \frac{4}{s^2}}\right\}, \tag{2.39}$$

with

$$t = \left(\frac{30.67}{s}\right)^{0.75}. \tag{2.40}$$

Figure 2-11 displays plots of Z_0 as a function of s for various types of dielectric materials.

The corresponding line and propagation parameters are given by

$$R' = 0 \quad \text{(because } \sigma_c = \infty\text{)}, \tag{2.41a}$$

$$G' = 0 \quad \text{(because } \sigma = 0\text{)}, \tag{2.41b}$$

$$C' = \frac{\sqrt{\varepsilon_{eff}}}{Z_0 c}, \tag{2.41c}$$

$$L' = Z_0^2 C', \tag{2.41d}$$

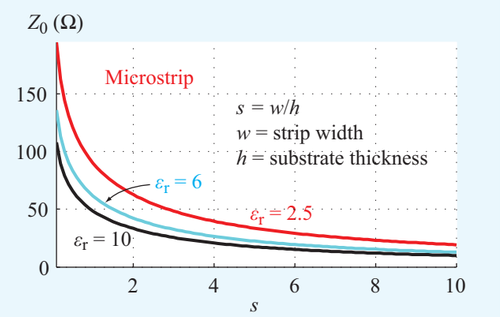


Figure 2-11 Plots of Z_0 as a function of s for various types of dielectric materials.

*D. H. Schrader, *Microstrip Circuit Analysis*, Prentice Hall, 1995, pp. 31–32.

实践中总是如此。此外，我们假设基板材料是σ = 0 的理想电介质，金属条和接地平面是σc ≈ ∞ 的理想导体。这两个假设大大简化了分析，而不会产生重大错误。最后，我们设置 μ = μ0，这对于微带线制造中使用的介电材料始终成立。这些简化将几何和材料参数的数量减少到三个，即 w、h 和 ε。

电场线总是从带正电荷的导体开始，到带负电荷的导体结束。对于图 2-4 上半部分所示的同轴、双线和平行板线，场线被限制在导体之间的区域。这种传输线的一个特征是沿着其中任何一条传播的波的相速度由下式给出

$$u_p = \frac{c}{\sqrt{\varepsilon_r}}, \tag{2.36a}$$

其中 c 是自由空间中的光速，εr 是导体之间介电介质的相对介电常数。

2-5.1 有效介电常数

在微带线中，尽管大多数将带状线连接到地平面的电场线直接穿过介电基板，但也有少数电场线同时穿过带状线上方的空气区域和介电层（图2-10（b））。这种不均匀的混合物可以通过定义有效相对介电常数 εeff 来解释，这样相速度就可以由类似于方程 1 的表达式给出。(2.36a)，即

$$u_p = \frac{c}{\sqrt{\varepsilon_{eff}}}. \tag{2.36b}$$

计算微带线传播特性的方法相当复杂，超出了本文的范围。然而，可以使用严格解的曲线拟合近似来得到以下表达式集：*

$$\varepsilon_{eff} = \frac{\varepsilon_r + 1}{2} + \left(\frac{\varepsilon_r - 1}{2}\right)\left(1 + \frac{10}{s}\right)^{-xy}, \tag{2.37a}$$

其中 s 是宽厚比，

$$s = \frac{w}{h}, \tag{2.37b}$$

x 和 y 是由下式给出的中间变量

$$x = 0.56\left[\frac{\varepsilon_r - 0.9}{\varepsilon_r + 3}\right]^{0.05}, \tag{2.38a}$$

$$y = 1 + 0.02\ln\left(\frac{s^4 + 3.7\times10^{-4}s^2}{s^4 + 0.43}\right) + 0.05\ln(1 + 1.7\times10^{-4}s^3). \tag{2.38b}$$

2-5.2 特性阻抗

微带线的特性阻抗由 b 给出 y

$$Z_0 = \frac{60}{\sqrt{\varepsilon_{eff}}}\ln\left\{\frac{6 + (2\pi - 6)e^{-t}}{s} + \sqrt{1 + \frac{4}{s^2}}\right\}, \tag{2.39}$$

和

$$t = \left(\frac{30.67}{s}\right)^{0.75}. \tag{2.40}$$

图 2-11 显示了各种类型介电材料的 Z_0 与 s 的函数关系图。

相应的线路和传播参数由下式给出

$$R' = 0 \quad \text{(because } \sigma_c = \infty\text{)}, \tag{2.41a}$$

$$G' = 0 \quad \text{(because } \sigma = 0\text{)}, \tag{2.41b}$$

$$C' = \frac{\sqrt{\varepsilon_{eff}}}{Z_0 c}, \tag{2.41c}$$

$$L' = Z_0^2 C', \tag{2.41d}$$

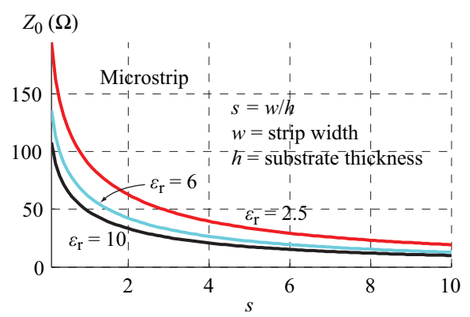


图 2-11 对于各种类型的介电材料，Z_0 与 s 的函数关系图。

*D. H. Schrader, *Microstrip Circuit Analysis*, Prentice Hall, 1995, pp. 31–32.